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Machine-learning dissection of endogenous A-to-I RNA editing reveals substrate- and target-level determinants of predictability

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Endogenous adenosine-to-inosine RNA editing depends on the sequence and structure of an RNA duplex, its availability to ADAR in a cell, and selection of a particular adenosine. Which measurements and modeling choices explain variation beyond current structure-aware predictors remains unclear. We used pair-disjoint machine-learning comparisons to test biological context, supervision, representation, transfer, and integration across 884 human inverted-Alu substrates while keeping the published test sealed. Pair geometry explained reproducible variation in substrate-level editing burden beyond graph-derived summaries, consistent with differences in realization or engagement, but current bulk cellular annotations did not resolve that state well enough to improve site prediction. Two changes consistently helped. Sampling substrates before sites corrected the disproportionate influence of site-rich duplexes, independently of the benefit from additional high- and low-editing observations. Aligning a local encoder on the target adenosine and its helix partner recovered context diluted by whole-duplex pooling. ADARPro combines independently trained whole-duplex-only and local-augmented graph predictors by a fixed equal-logit mean. Frozen before test access, it improved AUROC, AUPRC, and F1 in all five reported benchmark endpoints; mean gains were 0.0277, 0.0261, and 0.0253, with all 95% pair-clustered intervals above zero. Current measurements therefore capture intrinsic substrate suitability and target-local selectivity, whereas the cellular state governing substrate availability will require finer, matched measurements.

Significance

Why one apparently suitable adenosine is edited while another is not remains unclear. Using pair-disjoint machine-learning comparisons, we separated effects associated with the RNA duplex, substrate-level editing propensity, and the local environment of the target nucleotide. Pair geometry tracked editing burden, but tissue-average annotations did not resolve the underlying cellular state. Learning equally from RNA substrates and aligning each target with its helix partner consistently improved prediction. These results show how biology and statistical design jointly shape what can be learned from RNA-editing data and identify the matched cellular measurements needed next.

A-to-I RNA editing | ADAR | inverted Alu | RNA structure | machine learning

Adenosine deaminases acting on RNA (ADARs) convert adenosine to inosine in double-stranded RNA (dsRNA), diversifying transcripts while preventing endogenous duplexes from activating antiviral pathways (1–5). Most detected human editing occurs in paired Alu elements, whose abundance and orientation create imperfect intramolecular duplexes (6–10). Nevertheless, editing is selective: nearby adenosines within the same predicted structure can differ sharply, and the same locus can vary across tissues (11, 12). Local sequence preference, mismatches, bulges, and helix context contribute to this selectivity (13–17), but they do not specify whether a potential substrate forms, persists, localizes appropriately, or encounters the relevant ADAR state.

Recent structure-aware predictors, exemplified by AdarEdit, model intrinsic editability over complete inverted-Alu substrates (18). Its compact graph attention network encodes nucleotide identity, local and global sequence, predicted secondary structure, pairing topology, geometry, thermodynamics, and motifs. A strict benchmark assigns entire Alu pairs globally to train, validation, or test, avoiding the severe leakage that would occur if sites from one duplex crossed partitions. Residual errors could therefore reflect missing

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cellular context, underused supervision, pooling that dilutes the focal nucleotide, or an inadequate separation of substrate-level propensity from local selectivity.

Adding another annotation or a larger network cannot distinguish these explanations. A biologically relevant variable may be redundant with sequence, measured at the wrong cellular scale, or too noisy to improve prediction. An apparent architectural gain may instead arise because substrates with many labeled sites dominate training. We therefore asked: within measurements available to this benchmark, where does the remaining predictability of endogenous A-to-I editing reside?

We preserved the published benchmark and compared biological measurements, supervision schemes, substrate-wide and target-local encoders, cross-tissue transfer, RNA foundation models, hierarchical models, and two forms of integration. Each comparison asked whether a specific biological scale or modeling choice added to the sequence–structure graph. All development analyses used pair-disjoint folds and pair-clustered uncertainty; the canonical test remained sealed until the final predictor was frozen. Four conclusions emerged. Whole-duplex sequence and structure encode intrinsic substrate suitability. Editing burden also varies reproducibly among substrates in a manner consistent with realization or engagement, although current bulk measurements do not resolve the underlying cellular state. A target and its helix partner retain local context that whole-duplex pooling loses. Finally, the duplex is the effective statistical unit for learning from correlated sites. ADARPro was built from these findings rather than from indiscriminate increases in model capacity.

Results

Pair-disjoint comparisons separate biological scales. The benchmark comprises four individual GTEx tissue endpoints—artery tibial, brain/cerebellum, liver, and skeletal muscle—and Combined, a pooled endpoint rather than a fifth biological tissue. EditingIndex values at least 15% define positives and values below 1% define negatives; intermediate observations are excluded from the canonical binary task. The fixed global split contains 566 training, 139 validation, and 179 test pairs with zero overlap (Fig. 1).

Within the training and validation pairs, 5 outer folds generated out-of-fold (OOF) predictions for hypothesis testing and model selection. Feature fitting, checkpoint selection, and threshold choice were isolated from each outer fold. The internal reference was always trained and evaluated under the same folds as its candidate. Its lower absolute score reflects reduced training data and is not a replacement for published AdarEdit performance. The supplied-checkpoint canonical test remained the authoritative benchmark and was accessed only after the final model, thresholds, weights, code, and hashes had been frozen.

Intrinsic duplex properties and cell- or transcript-specific state jointly govern whether a substrate is available to editing; local sequence and pairing then help distinguish adenosines within it (Fig. 1A). We distinguished changes that improved site prediction from reproducible observations not yet useful to the classifier, smaller or context-specific effects, unsupported formulations, and questions that lacked matched measurements (Fig. 2A). A biologically relevant effect need not improve prediction at the resolution tested here.

The substrate is a biological and statistical unit. The canonical row set samples a subset of the available observations, whereas the underlying training tables contain additional sites with valid high or low EditingIndex. Recovering these labels improves supervision, but uniform site sampling gives a substrate weight proportional to its number of observations. Sites within one duplex share sequence, structure, genomic context, and often editing behavior; they are not independent biological units.

A same-fold ablation separated label recovery from pair weighting in muscle (Fig. 3A; SI Appendix). Adding all high/low observations under row-wise sampling improved AUROC by +2.04 percentage points (95% pair-clustered interval +1.06 to +3.00) and F1 by +1.42 points (+0.21 to +2.57). Holding that expanded pool fixed and drawing a pair before class and site contributed a further +2.27 AUROC points (+1.25 to +3.36) and +2.85 F1 points (+1.68 to +4.01). The two corrections therefore make separable contributions.

The combined pair-first formulation improved the same-fold reference in all five endpoints (Fig. 3B). All 15 AUROC, AUPRC, and F1 intervals excluded zero, with F1 gains from +0.69 to +4.23 percentage points. Pair-first sampling therefore improves learning by correcting the effective contribution of correlated sites.

The substrate also carries biological variation in editing burden. Genomic separation between Alu arms—information discarded when model inputs join them with a fixed separator—correlated negatively with edited-site fraction in every individual tissue (Spearman $\rho = -0.334$ to -0.482; Fig. 2B). Low-capacity models combining geometry with graph summaries predicted pair-level edited fraction with $R^2 = 0.050$–0.277 and outperformed graph summaries alone in every tissue (Fig. 2C; SI Appendix). Pair geometry thus identifies a reproducible substrate-level component of editing burden, consistent with variation in realization or engagement but not a direct measurement of either process.

Current bulk context measurements do not resolve the latent cellular state. Predicting editing burden at the pair level was easier than using it to classify individual sites. A cross-fitted engagement gate added 0.0084 AUPRC but only 0.0041 F1, whose interval included zero, and increased Brier score by 0.0051. A neural engagement auxiliary likewise failed its prespecified criterion. Static geometry marks a stable propensity; it cannot reveal whether a particular transcript formed the expected duplex, remained accessible, occupied the right compartment, or encountered active ADAR.

The initial cellular-context hypothesis focused on RBP competition. We combined replicated ENCODE eCLIP occupancy, PrismNet and RBPNet predictions, literature-prioritized regulators, GTEx tissue abundance, binding-by-abundance interactions, and randomized or shuffled controls (19–22). Alu multimapping makes a missing eCLIP peak ambiguous; sequence-based predictors supplied a complementary test independent of peak absence. Neither cell-line occupancy nor tissue-average abundance contributed reproducible residual information.

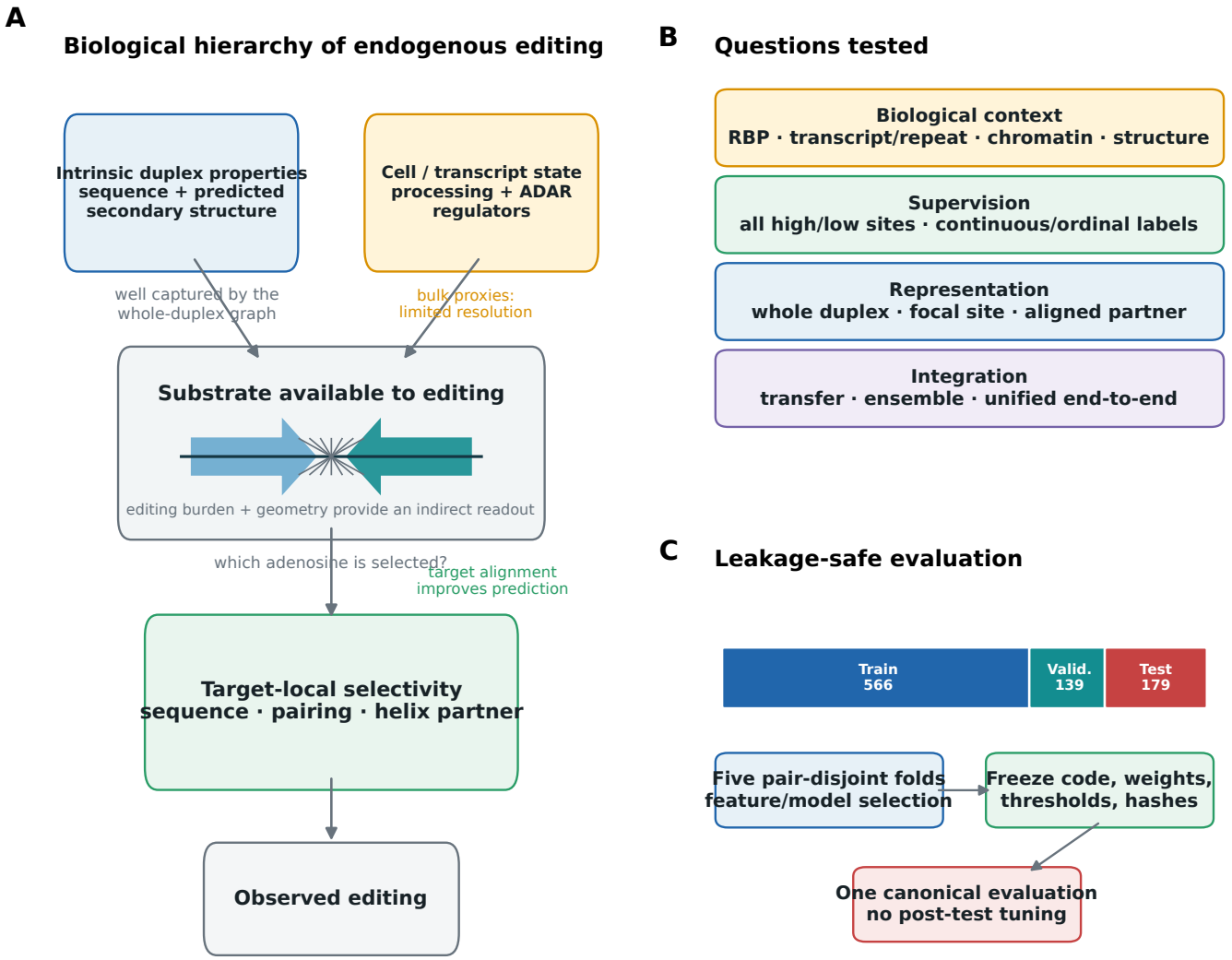


Fig. 1. Biological scales and leakage-safe machine-learning comparisons. (A) Intrinsic duplex properties and cell/transcript state converge on whether an inverted-Alu substrate is available to editing; target/helix-partner context then contributes to selection of a particular adenosine. Pair-level editing burden provides an indirect readout of substrate propensity, whereas the underlying cellular state was not measured directly. (B) Comparisons spanned biological context, supervision, representation, and integration. (C) Whole pairs were partitioned globally. All development work used pair-disjoint folds; the final specification was frozen before one canonical held-out evaluation. The pooled Combined endpoint received no tissue-specific state vector.

Cross-tissue label switches were also nearly separated by tissue identity alone, preventing four tissue means from identifying an RBP-specific transferable mechanism. These measurements are therefore unlikely to resolve transcript-specific regulator exposure; matched cell- or donor-level assays are the more informative next test (23–26).

Broad transcript, repeat-subfamily, divergence, and neighborhood annotations likewise failed to improve site-level prediction beyond the graph. Exact-tissue bulk H3K36me3 overlap produced a small liver AUPRC increment without stable AUROC, F1, or pair-level benefit; no adequately matched H2A.Z resource supported a powered test. The weak bulk H3K36me3 effect points toward quantitative, sample-matched assays of cotranscriptional state rather than binary tissue-level overlap (27, 28). Together, these results sharpen the measurement problem: substrate-level propensity is detectable, but available static and tissue-average proxies do not resolve the cellular state needed to improve the strongest site classifier.

Target alignment resolves site-level selectivity. The compact AdarEdit graph captures whole-duplex sequence and structural suitability well, but averaging nucleotide embeddings can dilute the state of one target. Two intermediate experiments localized the missing context. Simple target-contact geometry improved ranking and calibration but not F1, and explicitly retaining the focal graph node improved AUROC and AUPRC more consistently than an otherwise matched duplicate-global control. Both results implicated target position before a specialized local encoder was built.

The selected local-augmented graph predictor adds a 65-position representation centered jointly on the focal adenosine and its dot-bracket-aligned helix partner. A low-capacity dilated convolution encodes nucleotide identity, pairing state, focal and partner indicators, and signed relative position; its output is fused with the whole-duplex graph embedding before classification. In muscle OOF data it reached AUROC/AUPRC/F1 of 0.9045/0.8936/0.8337, compared

A

What adds to the sequence-structure graph?

Tested formulation	Evidence	Main finding
LEARNING & SUPERVISION		
Pair-first expanded supervision	Selected	More labels; correct pair weight
Continuous + ordinal auxiliary labels	Limited	Small ranking gain
Shared four-tissue training	Limited	Transfer below preset criteria
TARGET REPRESENTATION		
Aligned target + helix-partner view	Selected	Passes shifted-window control
Focal node / contact geometry	Limited	Localizes target effect
Frozen RNA-language embeddings	Not supported	F1-only; no ranking gain
Substrate-wide all-A decoder	Not supported	Stopped for futility
Larger / deeper graph networks	Not supported	No consistent benefit
Graph Transformer / GATv2 expansion	Not pursued	Simpler validated model sufficient
SUBSTRATE & CELLULAR CONTEXT		
Pair geometry / editing burden	Supported*	Substrate propensity detectable
Hierarchical engagement gate	Limited	Ranking gain; unstable F1
Bulk H3K36me3 overlap	Limited	Liver AUPRC only
RBP binding × bulk expression	Not supported	No residual gain
Transcript / repeat annotation panel	Not supported	No site-level gain
Matched donor/cell multimodality	Deferred	Matched data unavailable
Matched in vivo structure / isoforms	Deferred	Matched data unavailable
INTEGRATION		
Fixed whole-duplex + local-augmented logits	Selected	Complementary predictions
Unified end-to-end architecture	Not supported	Below frozen ensemble

See Table S2 for comparators, effect sizes, and intervals.
*Reproducible observation not yet useful for site prediction; deferred hypotheses were not tested.

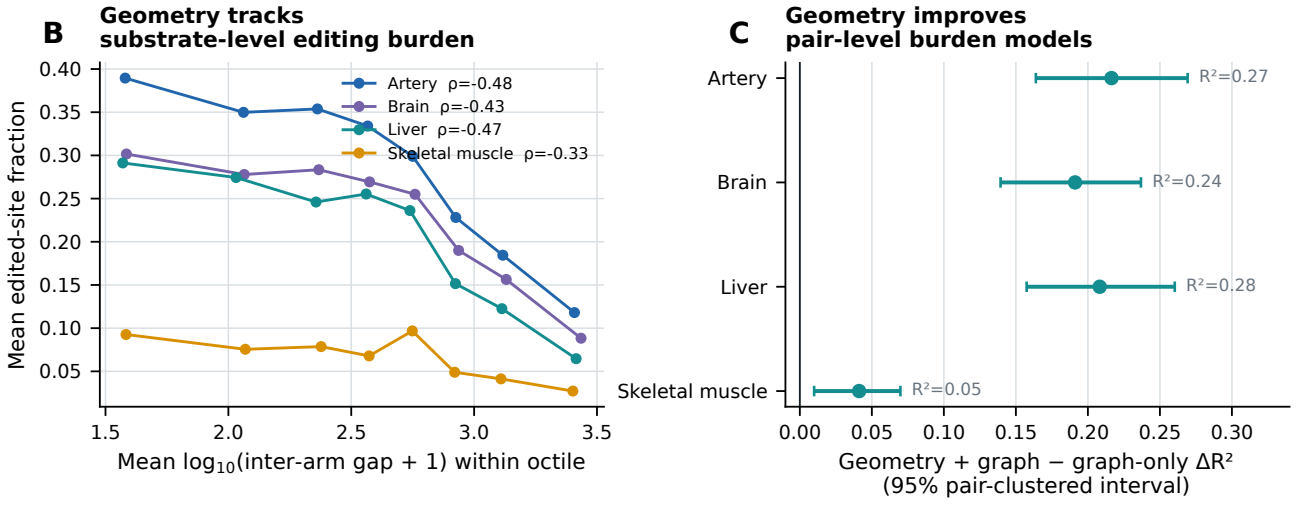


Fig. 2. What adds to the sequence–structure graph. (A) Hypotheses are grouped by domain and outcome; SI Appendix, Table S2 gives each comparator, effect size, and interval. "Not supported" refers only to the tested formulation, deferred measurements were unavailable, and superseded branches were not interpreted as negative. (B) Mean edited-site fraction across octiles of inter-arm gap in four tissues; ρ is calculated over individual development pairs. (C) Increment in pair-level R^2 from adding genomic geometry to graph-derived summaries. Bars are 95% pair-clustered bootstrap intervals; labels give absolute R^2 for the combined model.

with 0.8885/0.8730/0.8208 for the pair-first whole-duplex-only predictor (Fig. 3C).

The decisive control retained the same graph-plus-convolution architecture and optimization but shifted the local center by 96 nt. Performance fell to 0.7122/0.6943/0.7028. Aligned-minus-shifted differences were 0.1926 AUROC, 0.1993 AUPRC, and 0.1311 F1, with entirely positive pair-clustered intervals. This matched-capacity shift sharply reduced performance, localizing the gain to target alignment rather than added convolutional capacity. Liver replication and the secondary natural-prevalence analysis showed the same pattern.

Whole-duplex and target-local views are complementary. The local-augmented predictor did not make an independently optimized whole-duplex view redundant. Their errors remained sufficiently distinct that a fixed mean of logits improved the whole-duplex-only predictor by 0.0256/0.0308/0.0139 AUROC/AUPRC/F1 in muscle and 0.0182/0.0214/0.0179 in liver; all six pair-clustered intervals excluded zero (Fig. 3D).

ADARPro was consequently specified as two independently trained pair-first components: a whole-duplex-only graph predictor and a second graph predictor internally augmented with the aligned target/helix-partner convolution. Their logits are combined by a fixed 0.5/0.5 mean; the weights were not searched (Fig. 3E). Fixed integration preserved their complementarity without fitting a flexible stacker to only hundreds of independent substrates.

Additional complexity provides limited returns. Intermediate EditingIndex values contain graded information, but most useful thresholded content was already recovered by pair-first binary training. A binary/ordinal/continuous multitask head increased AUROC by 0.0081 (95% pair-clustered interval 0.0010–0.0151) while changing F1 by only 0.0008; its integration was adverse under natural prevalence. Shared four-tissue training likewise showed transferable rules, with mean changes across the four individual tissue endpoints of 0.0087 AUROC, 0.0114 AUPRC, and 0.0075 F1, but fell below its predeclared selection criteria and worsened mean Brier score by 0.0043. Graded labels and cross-tissue sharing therefore refine ranking, but less than correcting supervision and target representation.

Other increases in capacity were unproductive. A substrate-wide decoder trained all candidate adenosines jointly changed AUROC by -0.0664 and F1 by -0.0599 relative to the pair-first model and met its prespecified two-fold futility boundary. Frozen RiNALMo target and substrate embeddings, including a larger model, added little beyond the task-specific structure graph (29). Larger or deeper graph networks likewise failed stronger-model controls. More parameters did not compensate for misaligned supervision or representation. The sharper pretraining question—benchmark-safe, inverted-Alu-specific learning—remained deferred because a leakage-safe corpus could not be guaranteed.

Before test access, we preregistered one connected architecture to ask whether the successful components would interact more effectively during joint optimization. It combined a three-layer graph encoder, focal-node state, aligned local encoder, pairwise representation interactions, and a training-only substrate head. Its 65,698 trainable parameters were fewer than the 80,138 parameters across the two ensemble components. On the 5,967-row, 585-pair muscle development set, the unified model reached 0.9072/0.8922/0.8317 AUROC/AUPRC/F1 versus 0.9142/0.9040/0.8347 for the frozen ensemble. AUROC and AUPRC were worse in every fold; their pair-clustered intervals were wholly adverse, and Brier increased by 0.0205 (0.0135–0.0276). Under the preregistered decision rule, repair and replication stages were not opened. The tested joint architecture therefore did not outperform fixed prediction-level integration at this effective sample size.

The frozen predictor improves every canonical endpoint. After development selection, 10 component fits were trained on published training pairs. Published validation data alone selected component checkpoints and epochs and one ensemble threshold per endpoint. The exact code revision, configurations, weights, and hashes were bound into a final manifest. A single authorization then exposed the canonical test for one evaluation; no model or claim-defining analysis was changed afterward.

ADARPro improved AUROC, AUPRC, and F1 in all five reported endpoints (Fig. 4A). Mean gains across those endpoints over reproduced supplied-checkpoint AdarEdit were 0.0277, 0.0261, and 0.0253. Because Combined overlaps the individual tissue endpoints, this mean is descriptive rather than a five-tissue meta-analysis. F1 changed from 0.8671 to 0.8869 in artery, 0.8689 to 0.8902 in brain, 0.8541 to 0.8812 in liver, 0.8139 to 0.8541 in skeletal muscle, and 0.8674 to 0.8852 in Combined. Every endpoint had a positive 95% pair-clustered bootstrap interval for all three metrics, and Brier score decreased throughout (Fig. 4C).

The gains were not purchased by indiscriminate suppression of positive calls. At frozen validation-selected thresholds, ADARPro corrected 32.0–50.6% of baseline false positives while damaging 1.7–5.9% of baseline true positives (Fig. 4D). In a post-test descriptive analysis that rethresholded ADARPro to match AdarEdit recall exactly, the point estimates were 126, 119, 158, 68, and 148 fewer false positives across artery, brain, liver, skeletal muscle, and Combined; every pair-clustered interval excluded zero (Fig. 4E).

Discussion

The resulting picture describes what current data can predict, not a complete causal decomposition of RNA-editing biology. AdarEdit already captures substantial sequence- and structure-encoded substrate suitability. The same benchmark nevertheless contains an additional substrate-level dimension: inter-arm geometry covaries with editing burden, and low-capacity pair models predict that burden beyond graph-only summaries. Within a plausible substrate, the fate of an adenosine retains target/helix-partner context that global pooling dilutes. Sites sharing a duplex also form a correlated statistical unit; learning improves when substrates, rather than abundant rows, control effective weight.

The statistical and biological results answer different questions. Pair-first sampling corrects the weight of correlated sites; it does not require every site in a substrate to share one causal fate. Pair geometry, by contrast, identifies a stable component of substrate-level editing burden even though adding it did not improve the leading site classifier. Actual realization and engagement also depend on transcription, splicing, residence time, localization, in vivo conformation, ADAR isoform and activity, and regulator occupancy—states averaged away by tissue medians and binary peaks.

The branches that fell short make the remaining measurement gap more precise. Cell-line RBP occupancy combined with tissue-median abundance did not improve prediction beyond the graph and cannot report regulator exposure on individual transcripts. The weak bulk H3K36me3 effect makes quantitative, sample-matched cotranscriptional measurements the sharper next test. Cross-tissue sharing and continuous labels yielded small ranking gains, showing that transferable and graded structure exists but contributes less than corrected supervision and target alignment. Generic frozen RNA embeddings and greater graph capacity likewise did not recover the target-specific context missing from global pooling.

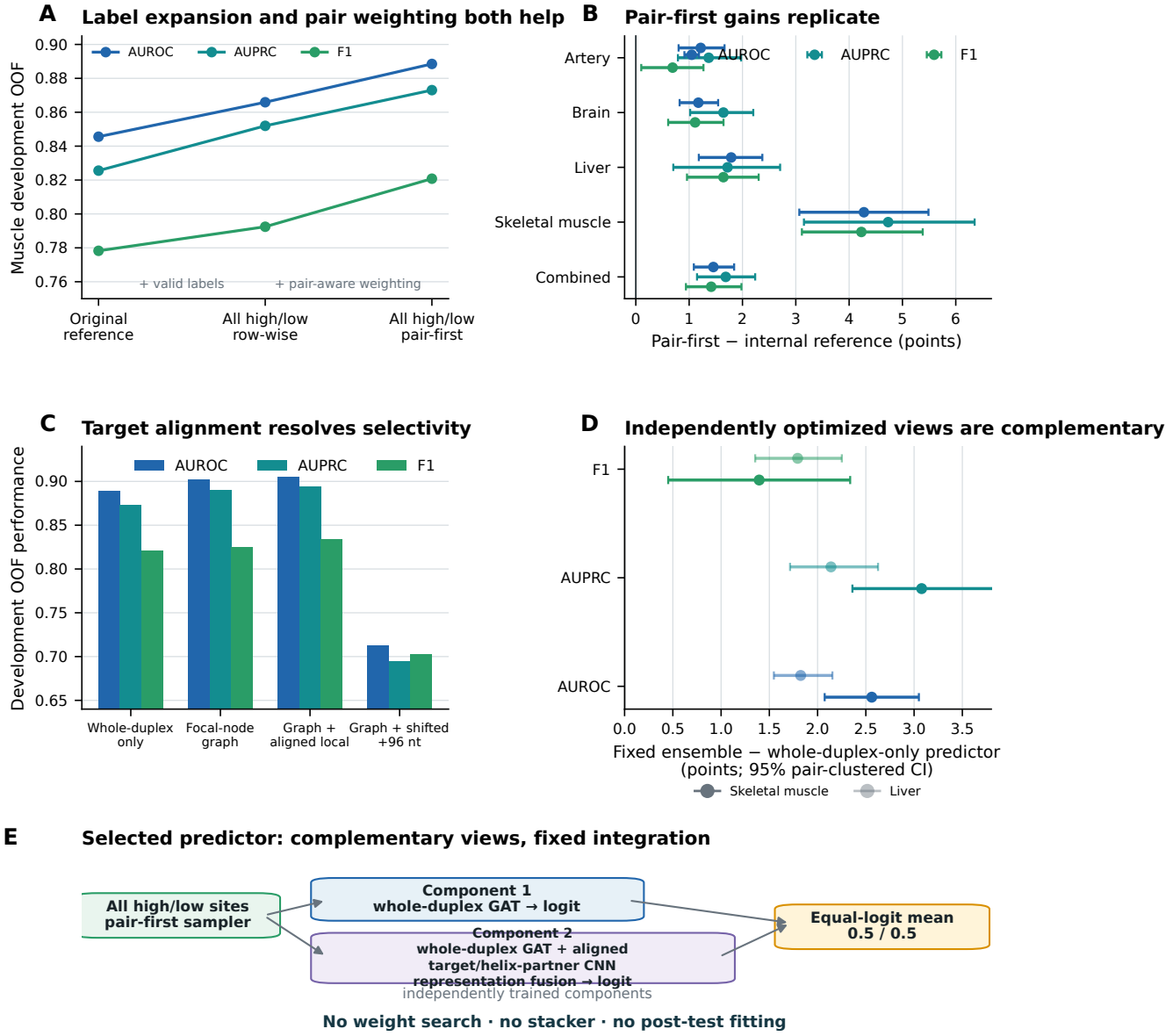


Fig. 3. Two machine-learning changes improve prediction and combine productively. (A) Same-fold ablations separate the benefit of recovering additional valid high/low labels under row-wise sampling from the further benefit of pair-first weighting. (B) Pair-first expanded supervision improves the internal reference across all five endpoints; points and bars are pair-clustered medians and 95% intervals. (C) Whole-duplex pooling, explicit focal-state retention, and graph-plus-aligned-local fusion form an evidence progression; an otherwise identical local-augmented model shifted by 96 nt fails. (D) The fixed equal-logit ensemble improves its whole-duplex-only component in muscle and liver. (E) The selected predictor consists of an independently trained whole-duplex-only graph model and graph-plus-local fusion model, followed by a fixed 0.5/0.5 logit mean. No ensemble weight, stacker, or post-test parameter is fitted. All panels show development evidence.

The preregistered unified experiment qualifies another common assumption: representation-level integration is not automatically superior to fixed prediction-level integration. The tested connected architecture ranked and calibrated less well than the frozen ensemble, despite incorporating the validated views and using fewer total parameters. The comparison does not identify the optimization mechanism. It does establish why the final predictor is deliberately simple: a whole-duplex-only graph and an independently trained graph-plus-local model, combined without a learned weight.

Several limitations define the scope of the conclusions. The benchmark contains graph-buildable human inverted-Alu substrates and tissue summaries, not donor-resolved measurements or all editing substrates. Predicted secondary structure is a surrogate for the conformations available to ADAR in vivo. The canonical task excludes intermediate observations and is balanced; natural-prevalence analyses are secondary. eCLIP coverage is cell-line-specific and compromised by repeat mapping. Finally, the canonical partition is stringent but derives from the same benchmark construction; validation in an independent cohort remains necessary.

Better prediction followed from respecting substrate dependence and preserving target-centered context, not from indiscriminate feature or capacity growth. The canonical gain provides an out-of-sample test of those development findings. The major unresolved layer is the

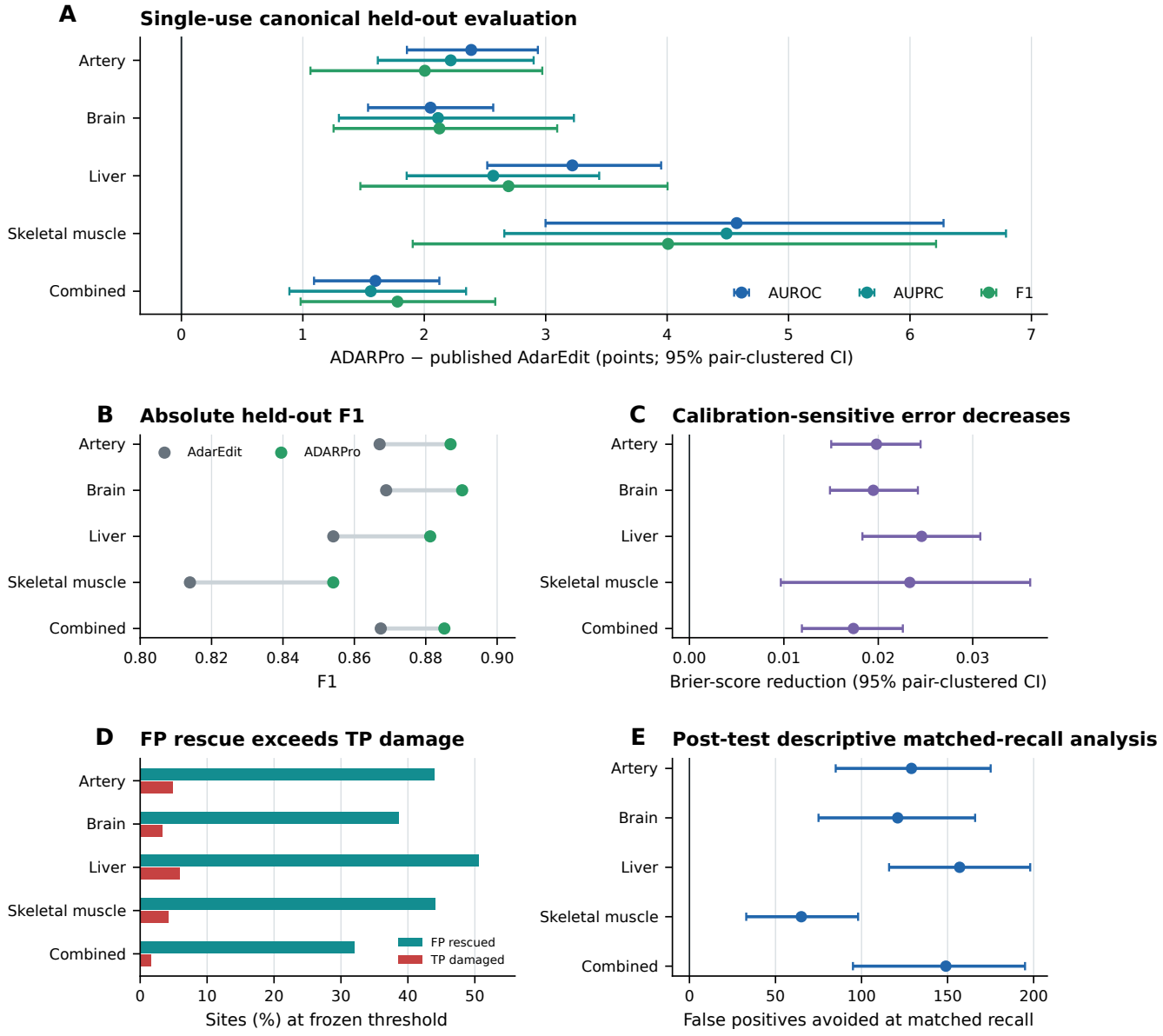


Fig. 4. Single-use canonical held-out evaluation. (A) ADARPro minus reproduced supplied-checkpoint AdarEdit for AUROC, AUPRC, and F1; bars are 95% pair-clustered bootstrap intervals. (B) Absolute F1. (C) Reduction in Brier score, a calibration-sensitive error measure. (D) Baseline false positives corrected and baseline true positives damaged at frozen validation-selected thresholds. (E) Descriptive post-test false-positive reduction after rethresholding ADARPro to match AdarEdit recall. Combined is a pooled endpoint. All quantities were generated after the final manifest was frozen and the one-use authorization was consumed.

cellular state underlying substrate-level propensity. The next decisive study should measure editing, transcript abundance and processing, splicing and intron retention, ADAR1 isoform and interferon state, ADAR activity and localization, regulator occupancy, quantitative chromatin, and in vivo RNA conformation in the same donors or cells and on the same substrates. Such data could directly test a two-stage model in which cellular state governs whether an intrinsically suitable substrate is realized or engaged, and local RNA context governs which adenosine is selected.

Materials and Methods

Benchmark and labels. We used the AdarEdit v2 reproducibility release at upstream revision `5c8b57d70cdc2eb1c5e168431013275bcc9ed0c5`. The benchmark contains 884 graph-buildable inverted-Alu pairs. Human labels were edited for EditingIndex $\geq 15\%$, nonedited for EditingIndex $< 1\%$, and intermediate otherwise. The canonical task excludes intermediate observations. Pair identities and the fixed 566/139/179 train/validation/test assignment were verified before analysis. Combined was treated only as its published pooled endpoint.

Development protocol. All discovery used the original train plus validation pairs divided into 5 outer folds by pair ID. Learned preprocessing, feature selection, checkpoint choice, and threshold choice were fitted without the outer fold. Predictions were concatenated only after every row was scored by a model that had not trained on its pair. The test paths were protected by a stateful quarantine. Development OOF metrics estimate same-fold incremental effects; supplied-checkpoint test metrics remain the authoritative reference.

Models. The first component retained the compact AdarEdit graph attention encoder and trained from the expanded high/low site pool using uniform pair-first and equal-class sampling. The second, independently trained component retained the same graph encoder and added a 65-position

target/helix-partner-aligned representation encoded by two dilated convolutional layers; local and graph embeddings were fused before its prediction head. The selected probability was $\sigma[(\mathrm{logit}\, p_{\mathrm{whole}} + \mathrm{logit}\, p_{\mathrm{augmented}})/2]$; weights were fixed at 0.5/0.5 rather than searched. The shifted-window control preserved the graph-plus-convolution architecture and optimization while moving the local center 96 nt. Full feature definitions, architectures, ablations, and biological-context protocols are in SI Appendix.

Statistics and final evaluation. Metrics were AUROC, AUPRC, F1, precision, recall, specificity, balanced accuracy, MCC, and Brier score. Thresholds maximized validation F1 within each frozen training run. Uncertainty used a pair-clustered bootstrap with 2,000 replicates that resampled Alu pairs and retained all their sites. For the post-test descriptive matched-recall analysis, the candidate threshold was selected on final predictions solely to equal supplied-checkpoint baseline recall; uncertainty again resampled pairs. Before final evaluation, code, inputs, weights, epochs, thresholds, and checkpoint SHA-256 values were frozen. Test authorization was consumed once, and no post-test model selection or tuning occurred. Natural-prevalence results use all eligible high/low observations from development pairs under unchanged pair assignments and are secondary to the canonical balanced task.

Data, Materials, and Software Availability. Code, experiment configurations, registries, source tables, figure scripts, retained derived artifacts, frozen checkpoints, and provenance are available at https://github.com/Scientific-Computing-Lab/ADARPro. Third-party downloads and deterministic matrices omitted from GitHub are listed with source URLs, roles, checksums, and regeneration instructions in the data manifest. The AdarEdit benchmark is available from https://github.com/Scientific-Computing-Lab/AdarEdit. No new human participants were recruited and no new primary sequencing data were generated.

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